

Alessandro Baccarini

Contact Information

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📄 Scholar 1334 citations, Aug. 2026

Research Interests

My research interests span applied cryptography, information security, and privacy, with an emphasis on distributed cryptographic protocols, privacy-preserving computation, and blockchain-based systems. A central theme of my work is bridging theory and practice: I design secure and efficient systems and protocols that connect rigorous cryptographic foundations with real-world applications, particularly in decentralized and heterogeneous settings.

Education

PhD, Computer Science, University at Buffalo Aug. 2024
Advisor: Marina Blanton
MS, Cybersecurity, Fordham University May 2019
Advisor: Thaier Hayajneh
BS, Physics, Fordham University May 2017
Minor in Mathematics

Work Experience

Chief Cryptographer, Pogun June 2026 – Present
Input Output Group
Applied Cryptographer, Partner Chains, Pogun Sep. 2025 – June 2026
Input Output Group
Principal Consultant, Cryptography Sep. 2024 – Present
Guardian Cryptography
Research Assistant, Computer Science Jun. 2019 – Aug. 2024
University at Buffalo
Teaching Assistant, Computer Science Jan. 2020 – May 2022
University at Buffalo
Adjunct Assistant Professor, Physics Aug. 2017 – May 2019
Fordham University
Graduate Research Assistant, Cybersecurity Aug. 2017 – May 2019
Fordham University

Awards and Recognition

Distinguished Artifact Reviewer , Privacy Enhancing Technologies Symposium (PETS)	July 2026
Noteworthy Artifact Reviewer , USNEIX Security Symposium	Aug. 2026
Alan Selman Scholarship , University at Buffalo First place \$2000 cash prize, focus in theoretical computer science.	Mar. 2024
GSAS Centennial Scholarship , Fordham University Full tuition support and stipend (academic year + summer).	Aug. 2017 – May 2019

Experience

Pogun Bridge (Mirror) Input Output Group, <i>Pogun</i>	Dec. 2025 – Present
- Leading the cryptographic research, design, and execution of the <i>Pogun Mirror</i>, a trust-minimized bridge architecture between Bitcoin and Cardano based on SNARK witness encryption and secure multi-party computation (MPC). - Extended a novel passively (semi-honest) secure two-party bridge construction to support an arbitrary number of parties with active (malicious) security by incorporating Verifiable Shamir Secret Sharing (VSSS) and Schnorr Adaptor Signatures into the framework. - Implemented cryptographic protocols for the bridge prototype in Rust, demonstrated the efficiency and practicality of the construction, and delivered the prototype to the core engineering team for productionization. - Coordinating with third party organizations to arrange a comprehensive security audit of our proposed construction. - Actively researching techniques to improve overall efficiency (via multi-party garbling and distributed polynomial evaluation) and to harden the system against post-quantum adversaries.	
Capacity Exchange Threshold Signature Input Output Group, <i>Partner Chains</i>	Sep. 2025 – Dec. 2025
- Led the cryptographic design of the Capacity Exchange for the Midnight blockchain, a decentralized platform that enables externally funded private transactions (from Bitcoin, Ethereum, and Cardano) on Midnight by leveraging threshold signature schemes (TSS) and zero-knowledge proofs. - Evaluated state-of-the-art TSS constructions (FROST, MuSig2, and their post-quantum variants) based on their security guarantees, performance characteristics, and suitability for the Capacity Exchange ecosystem. - Collaborated closely with internal and external R&D teams to identify opportunities for novel cryptographic and economic research and to design bespoke engineering and architectural solutions.	
Principal Consultant Guardian Cryptography	Sep. 2024 – Present
Provide expert consulting in applied cryptography, privacy-enhancing technologies, and secure software development for startups, research teams, and enterprise clients.	
Threshold Decryption for FHE Guardian Cryptography. <i>Client: Blockchain R&D Organization</i>	Sep. 2024 – Dec. 2024
- Analyzed distributed threshold decryption protocols for multi-party fully homomorphic encryption (FHE) schemes with application to private smart contract deployment on Ethereum-like blockchains. - Implemented and evaluated an actively secure threshold decryption construction based on Shamir secret sharing over Galois rings in C++, obtaining an up to $4\times$ performance improvement over prior works.	

- Designed a general-purpose threshold distributed key generation protocol compatible with various multi-party FHE schemes, and implemented it within the MP-SPDZ framework.

PICCO Compiler

Jan. 2020 – Aug. 2024

University at Buffalo

[Repository](#)

- Core developer and maintainer of PICCO, a source-to-source MPC compiler library for translating general-purpose programs into their secure distributed equivalents.
- Integrated ring-based constructions into the compiler to support general-purpose secure computation over diverse input domains and broaden application flexibility.
- Performed extensive optimizations to existing field-based protocols and led a large-scale refactor of over 100k lines of code to enhance long-term maintainability and enable support for stronger security models.
- Mentored undergraduate students on projects including optimizing and parallelizing networking layers across parties, and developing a web interface to facilitate secure input/output interactions.

MPC Framework for Privacy-Preserving AI

Jan. 2020 – July 2023

University at Buffalo

[Repository](#)

- Designed a novel comprehensive ring-based framework of replicated secret sharing MPC protocols for an arbitrary number of parties in the semi-honest, honest majority setting.
- Implemented protocols in C++, applying extensive profiling and low-level optimizations that led to up to $33\times$ performance improvements over state-of-the-art secret sharing techniques.
- Applied MPC to privacy-preserving machine learning tasks, including (quantized) convolutional deep neural network inference and support vector machine classification.
- Discovered an algebraic optimization for secure quantized inference that minimizes the overall ring modulus across multiple layer evaluations, yielding over $2\times$ performance improvement on average.

Secure Computation Information Disclosure Analysis

Jan. 2021 – Present

University at Buffalo

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- Developed an information-theoretic technique to quantify leakage about private inputs from arbitrary secure computation outputs, enabling practical assessment of residual disclosure under complex function evaluation.
- Analyzed common statistical functions under practical MPC configurations and proposed concrete mitigation strategies for real-world deployment.
- Combined the methodology with entropy estimation techniques using machine learning to assess leakage from complex descriptive statistical measures (e.g., variance, order statistics).
- Built a toolkit written in C++ and Python to evaluate, measure, and report potential information-disclosure vectors from user-supplied secure computation programs.
- Awarded first place Alan Selman Scholarship in Theoretical Computer Science for this work, recognized for its combination of rigorous theory and real-world relevance to secure data analysis.

Publications

- [1] **Alessandro Baccarini**, Marina Blanton, and Shaofeng Zou. Understanding information disclosure from secure computation output: A study of average salary computation. In *ACM Conference on Data and Application Security and Privacy (CODASPY)*, pages 187–198, 2024.
- [2] **Alessandro Baccarini**. *New Directions in Secure Multi-Party Computation: Techniques and Information Disclosure Analysis*. PhD thesis, University at Buffalo, 2024.
- [3] **Alessandro Baccarini**, Marina Blanton, and Shaofeng Zou. Understanding information disclosure from secure computation output: A comprehensive study of average salary computation. *ACM Transactions on Privacy and Security (TOPS)*, 28(1):1–36, 2024.

- [4] **Alessandro Baccarini**, Marina Blanton, and Chen Yuan. Multi-party replicated secret sharing over a ring with applications to privacy-preserving machine learning. *Proceedings on Privacy Enhancing Technologies (PoPETs)*, 2023(1):608–626, 2023.
- [5] **Alessandro Baccarini** and Thaier Hayajneh. Evolution of format preserving encryption on IoT devices: FF1+. In *Hawaii International Conference on System Sciences (HICSS)*, pages 1628–1637, 2019.
- [6] Abdullah Alhayajneh, **Alessandro Baccarini**, Gary Weiss, Thaier Hayajneh, and Aydin Farajidavar. Biometric authentication and verification for medical cyber physical systems. *Electronics*, 7(12):436, 2018.
- [7] Abdullah Alhayajneh, **Alessandro Baccarini**, and Thaier Hayajneh. Quality of service analysis of VoIP services. In *IEEE Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON)*, pages 812–818, 2018.
- [8] Kristen Griggs, Olya Ossipova, Christopher Kohlios, **Alessandro Baccarini**, Emily Howson, and Thaier Hayajneh. Healthcare blockchain system using smart contracts for secure automated remote patient monitoring. *Journal of Medical Systems*, 42(7):130, 2018.

In Preparation

- [9] **Alessandro Baccarini**, Marina Blanton, and Adithya Vadapalli. Private data streaming statistics, 2026.
- [10] **Alessandro Baccarini**, Marina Blanton, and Shaofeng Zou. Simulation-based techniques for estimating information disclosure from secure function evaluations, 2026.

Technical Presentations

- Molekule Consulting, AI Security & Privacy. Virtual. Sep. 2025
- RAND Corporation, Engineering & Applied Sciences Dept. Virtual. Apr. 2025
- Intel Labs, Security & Privacy Research Group. Virtual. Mar. 2025
- Riverside Research, Secure & Resilient Systems Group. Lexington, MA. Feb. 2025
- MITRE Corporation, Cyber for Identity Trust & Assurance Dept. Virtual. Jan. 2025
- ACM CODASPY 2024. Porto, Portugal. June 2024
- PETS 2023. Lausanne, Switzerland. July 2023
- Great Lakes Security Day, RIT. Rochester, NY. Apr. 2023
- IEEE UEMCON 2018. New York, NY. Nov. 2018

Professional Service

Conference Committees

- Privacy Enhancing Technologies Symposium (PETS), artifact evaluation committee 2026
- USENIX Security Symposium, artifact evaluation committee 2023, 2024, 2026
- IEEE Symposium on Security and Privacy, poster jury 2025

Refereeing

- Journal of Computer and System Sciences (JCSS)
- IEEE Transactions on Information Forensics and Security (TIFS)
- IEEE Transactions on Dependable and Secure Computing (TDSC)
- European Symposium on Research in Computer Security (ESORICS)
- IEEE/ACM International Conference on Automated Software Engineering (ASE)

Technical Skills

Cryptographic	secure multi-party computation, secret sharing, witness encryption, threshold signature schemes, homomorphic encryption, lattice cryptography, zero-knowledge proofs, differential privacy, information theory
Languages	C/C++, Python, Rust, Bash, Lua, $\LaTeX$
Developer	Version control (Git, SVN), CMake, Make, GDB, Valgrind, Neovim, VS Code
Platforms	Docker, AWS EC2, GitHub, HPC Clustering, Linux, Unix, Windows
Libraries	arkworks, MP-SPDZ, GMP, GMPFR, GSL, STL, OpenSSL, SageMath, NumPy, Pandas, SciPy, Matplotlib, TensorFlow

Teaching

At the **University at Buffalo**:

CSE 116 Computer science II (Instructor)	2 semesters
CSE 4/529 Algorithms for Modern Computing Systems (TA)	3 semesters
CSE 4/531 Analysis of Algorithms (TA)	1 semester
CSE 542 Software Engineering Concepts (TA)	1 semester

At **Fordham University**:

PHYS 1511/12 Physics I/II Lab (Instructor)	4 semesters
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